

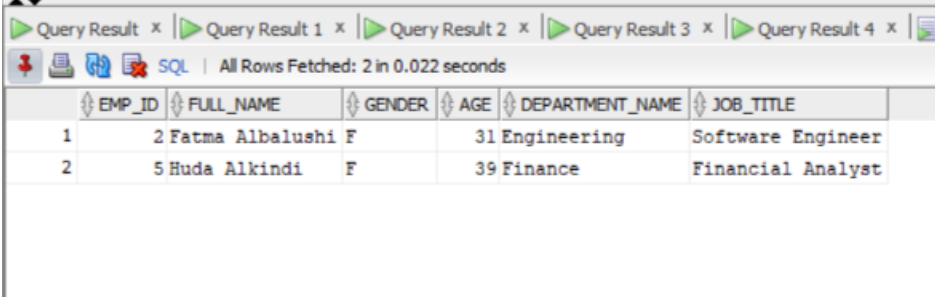
Section 07 — Views

Simple Read-Only View

```
CREATE OR REPLACE VIEW VW_EMPLOYEE_SUMMARY AS
SELECT e.emp_ID,
       e.fname || ' ' || e.lname AS full_name,
       e.gender,
       e.age,
       jd.name AS department_name,
       q.position AS job_title
FROM EMPLOYEE e
JOIN JOB_DEPARTMENT jd
ON e.job_ID = jd.job_ID
JOIN QUALIFICATION q
ON e.emp_ID = q.emp_ID
WITH READ ONLY;
```

Female Employees Older Than 30

```
SELECT *
FROM VW_EMPLOYEE_SUMMARY
WHERE gender = 'F'
AND age > 30;
```

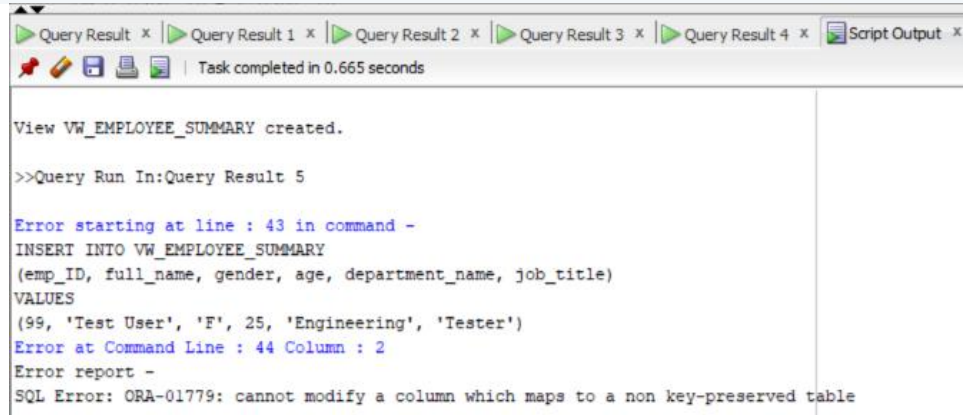


The screenshot shows a database query result window with the following data:

EMP_ID	FULL_NAME	GENDER	AGE	DEPARTMENT_NAME	JOB_TITLE
1	2 Fatma Albalushi	F	31	Engineering	Software Engineer
2	5 Huda Alkindi	F	39	Finance	Financial Analyst

INSERT into Read-Only View

```
INSERT INTO VW_EMPLOYEE_SUMMARY
(emp_ID, full_name, gender, age, department_name, job_title)
VALUES
(99, 'Test User', 'F', 25, 'Engineering', 'Tester');
```



The screenshot shows the Oracle SQL Developer interface. The top toolbar includes icons for running queries, saving, and other standard database operations. Below the toolbar, a status bar indicates "Task completed in 0.665 seconds". The main text area displays the following output:

```
View VW_EMPLOYEE_SUMMARY created.

>>Query Run In:Query Result 5

Error starting at line : 43 in command -
INSERT INTO VW_EMPLOYEE_SUMMARY
(emp_ID, full_name, gender, age, department_name, job_title)
VALUES
(99, 'Test User', 'F', 25, 'Engineering', 'Tester')
Error at Command Line : 44 Column : 2
Error report -
SQL Error: ORA-01779: cannot modify a column which maps to a non key-preserved table
```

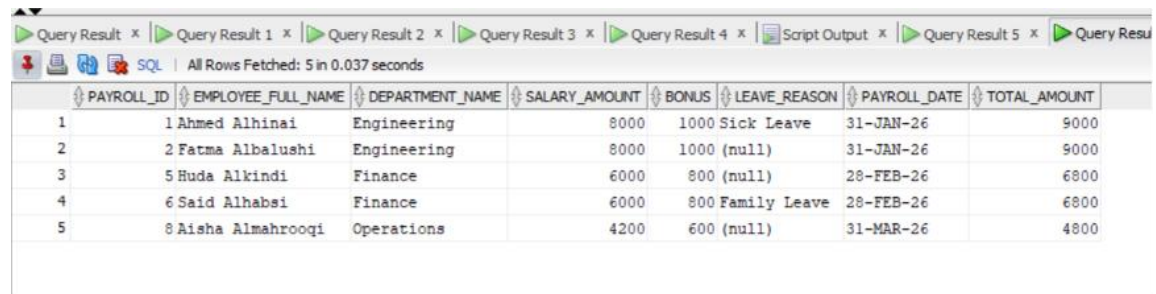
- The error happened because VW_EMPLOYEE_SUMMARY is based on joined tables so Oracle does not allow inserting data through this view

Payroll Dashboard View

```
CREATE OR REPLACE VIEW VW_PAYROLL_DASHBOARD AS
SELECT p.payroll_ID,
       e.fname || ' ' || e.lname AS employee_full_name,
       jd.name AS department_name,
       sb.amount AS salary_amount,
       sb.bonus,
       l.reason AS leave_reason,
       p.pay_date AS payroll_date,
       p.total_amount
FROM PAYROLL p
JOIN EMPLOYEE e
ON p.emp_ID = e.emp_ID
JOIN JOB_DEPARTMENT jd
ON p.job_ID = jd.job_ID
JOIN SALARY_BONUS sb
ON p.salary_ID = sb.salary_ID
LEFT JOIN LEAVE l
ON p.leave_ID = l.leave_ID;
```

Top 5 Payroll Records by Total Amount

```
SELECT *
FROM VW_PAYROLL_DASHBOARD
ORDER BY total_amount DESC
FETCH FIRST 5 ROWS ONLY;
```



The screenshot shows a SQL query result window with the following data:

PAYROLL_ID	EMPLOYEE_FULL_NAME	DEPARTMENT_NAME	SALARY_AMOUNT	BONUS	LEAVE_REASON	PAYROLL_DATE	TOTAL_AMOUNT
1	1 Ahmed Alhinai	Engineering	8000	1000	Sick Leave	31-JAN-26	9000
2	2 Fatma Albalushi	Engineering	8000	1000	(null)	31-JAN-26	9000
3	5 Huda Alkindi	Finance	6000	800	(null)	28-FEB-26	6800
4	6 Said Alhabsi	Finance	6000	800	Family Leave	28-FEB-26	6800
5	8 Aisha Almahrooqi	Operations	4200	600	(null)	31-MAR-26	4800